

with an intuitive touchscreen input," explains Seng.

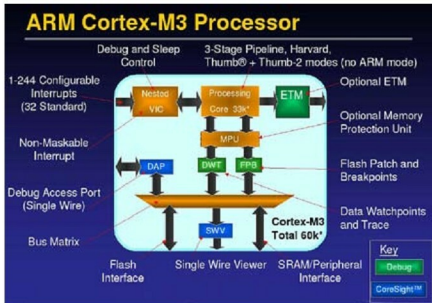
Portable medical devices such as blood pressure monitors need to provide more information about individual users. The ability to connect to a PC (wireless preferred), data processing and data analysis are some of the other features that are enhanced with the use of 32-bit MCUs. More processing power is required to handle multiple tasks.

Applications like industrial automation tools, medical equipments, point-of-sales terminals, digital weighing scales, printers, motor control and display have migrated from 8-bit/16-bit MCUs to 32-bit platform. "However, as 32-bit MCUs are now available at the price of 8-bit MCUs, soon you will see almost every application using 32-bit MCUs," says Ashok Chandak, senior director of global sales and marketing, NXP Semiconductors.

Advantage of 32-bit MCUs

The biggest challenge for the embedded manufacturers is the availability of many differing architectures and implementations from different MCU vendors. There is a need to support every such architecture used in the projects, which has reduced code reuse across projects and therefore the engineering efficiency. "This emphasises the need for a single common platform and high-level language," shares Seng. "Commercial and publicly available software such as the OS and middleware help to shorten development time. These portable software are targeted more towards 32-bit MCU and not the lower-performance MCUs."

The 32-bit MCU for an embedded system has two-fold advantage in terms of executing code. "The 32-bit MCU takes one clock cycle for execution of two bytes, which is twice as quick as compared to 8-bit and 16-bit. Fewer clock cycles means longer sleeping time and therefore energy saving, while smaller code size allows



Advantages of 32-bit MCUs over 8-bit

- High operating frequency
- Low active power requirement
- Code reuse
- Code size reduction by 40-50 per cent
- Enhanced features for increasingly complex algorithms
- Cost reduction due to device aggregation
- External connectivity—Ethernet, USB, CAN
- Efficient real-time operations
- Availability of tools

smaller memory size and therefore cost saving," says Chandak. "The code size is reduced by 40-50 per cent compared to 8-bit/16-bit MCUs," he adds.

The powerful architecture of 32-bit MCU enables it to deal with more

peripherals and memory. It is able to handle more tasks within the smaller embedded systems. Seng adds, "As the processing speed is adequate, 32-bit MCU is able to deal with more accurate data processing (which includes the floating-point data). This makes development and debugging much easier."

32-bit cores feature a variety of on-chip peripherals for communications, such as universal serial bus (USB), controller area network, Ethernet, as well as I²C interfaces and serial peripheral interface. Some also offer on-chip TFT/LCD drive capability, which reduces system costs for developers.

The increasing complexity of applications required many 8-bit systems to interconnect. With the advent of low-cost 32-bit MCUs, a single microcontroller can be used, reducing the overall costs of device aggregation.

Development in silicon technologies has significantly lowered the cost of 32-bit MCUs. However, the cost still depends on the pin count, flash size, peripherals, on-chip drivers, etc. "You will see a small-memory-size 32-bit MCU (16kB, 48-pin) costing less than a 16-bit MCU (256kB, 100-pin). In general, you may see a 32-bit MCU starting from a dollar plus, and also 8-/16-bit

A developer using 'C' language for programming needs to remember that all memory/data widths are 32-bit. The rest of the things are taken care by the compilers